TELCO CUSTOMER CHURN

Author: SRISHTI | Dataset: IBM Telco Customer Churn | Model: XGBoost

1. Business Problem

Customer churn is a significant challenge in the telecommunications industry. Retaining existing customers is substantially more cost-effective than acquiring new ones. The objective of this project is to:

- Identify patterns and factors associated with customer churn
- Detect high-risk customer segments for targeted intervention
- Build a predictive model capable of flagging likely churners before they leave
- Provide actionable business recommendations backed by statistical and model-driven evidence

The analysis uses a dataset of **7,043 telecom customers** containing demographic data, service subscriptions, billing behaviour, and contract details.

2. Dataset Overview

Source: IBM Telco Customer Churn Dataset
Shape: 7,043 rows × 21 columns
Target Variable: Churn (Yes / No)
Overall Churn Rate: ~26%

Feature Summary

Category	Features
Demographics	gender, SeniorCitizen, Partner, Dependents
Phone Services	PhoneService, MultipleLines
Internet Services	InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies
Account Info	tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
Target	Churn

3. Data Understanding & Cleaning

Initial Inspection

The following issues were identified and addressed:

Issue	Action Taken
TotalCharges stored as string	Converted to numeric using pd.to_numeric(errors='coerce')
11 customers with tenure = 0 had null TotalCharges	Filled with MonthlyCharges (reasonable for first billing cycle)
OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies had 'No internet service' values	Replaced with 'No' for consistency
MultipleLines had 'No phone service' values	Replaced with 'No'
SeniorCitizen was encoded as 0/1 while all other binary columns used Yes/No	Mapped to 'No'/'Yes' for uniformity
customerID is a non-informative identifier	Dropped from the dataset
Null values	None found (post-fix)
Duplicate rows	None found

Dataset After Cleaning

- **Shape:** 7,043 rows × 20 columns (after dropping customerID)
- **Null values:** 0
- **Duplicates:** 0
- A binary numeric version of the target, Churn_Binary (Yes=1, No=0), was created for use in statistical analysis and correlation plots.

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4. Univariate Analysis

Categorical Features

Key distributions observed across all 17 categorical columns:

Feature	Notable Distribution
Churn	~73% No, ~27% Yes — mild class imbalance
gender	Near equal split (~50% each)
SeniorCitizen	~84% Non-Senior, ~16% Senior

Feature	Notable Distribution
Partner	~48% have a partner
Dependents	~70% have no dependents
PhoneService	~90% use phone services
InternetService	~44% Fiber Optic, ~34% DSL, ~22% None
Contract	Month-to-month most popular (~55%), then Two Year (~24%), One Year (~21%)
PaymentMethod	Electronic check most common, then Mailed check, Bank transfer, Credit card
PaperlessBilling	~59% use paperless billing
OnlineSecurity	Only ~28% have it
TechSupport	Only ~29% have it
StreamingTV / StreamingMovies	~38–39% each

Imbalance note: A naive model predicting "No Churn" for every customer achieves ~73% accuracy. Accuracy alone is therefore a misleading metric. F1-Score, ROC-AUC, and Recall were used as primary evaluation criteria.

Numerical Features

Feature	Observation
tenure	Bimodal — peaks at 0–10 months and 65–70 months; slightly right-skewed
MonthlyCharges	Bimodal — peak at 20–30 and 70–100; slightly left-skewed
TotalCharges	Highly right-skewed; majority of customers have total charges under ₹1,000

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5. Bivariate Analysis

Categorical Features vs Churn

Each categorical feature was compared against churn rate using bar charts. A dashed reference line marks the overall average churn rate. Bars above indicate above-average churn risk.

Key findings:

Feature	Finding
Contract	Month-to-month customers churn significantly more; two-year customers churn least

Feature	Finding
PaymentMethod	Electronic check users have the highest churn; auto-pay methods show lower churn
InternetService	Fiber Optic customers churn far more than DSL customers
OnlineSecurity / TechSupport	Customers without these services churn more
OnlineBackup / DeviceProtection	Similar pattern — lack of services linked to higher churn
SeniorCitizen	Senior customers churn noticeably more
Partner / Dependents	Customers without a partner or dependents churn more
PaperlessBilling	Paperless billing users have higher churn (likely a proxy for other behaviours)
StreamingTV / StreamingMovies	Mild effect — slight uptick in churn for subscribers
gender	No meaningful difference — gender is a weak churn predictor
PhoneService	Very small difference — weak predictor

Numerical Features vs Churn

Box plots and KDE plots compared distributions of numerical features for churned vs non-churned customers.

Feature	Finding
tenure	Churned customers median ~10 months vs ~38 months for non-churned; strongest separator
MonthlyCharges	Churned median ~₹80 vs ~₹65 for non-churned; moderate overlap
TotalCharges	Churned much lower (shorter tenure); inversely related to churn

Conclusion: Low tenure + high monthly charges = highest churn risk profile for numerical features.

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6. Statistical Significance Testing

T-Tests (Numerical vs Churn)

Feature	Mean (Churned)	Mean (Not Churned)	p-value	Significant?
tenure	~18 months	~38 months	0.0000	✓ Yes

Feature	Mean (Churned)	Mean (Not Churned)	p-value	Significant?
MonthlyCharges	~₹74	~₹61	0.0000	✓ Yes
TotalCharges	~₹1,532	~₹2,553	0.0000	✓ Yes

All three numerical features showed statistically significant differences between churned and non-churned groups.

Chi-Square Tests + Cramér's V (Categorical vs Churn)

Cramér's V quantifies the strength of association on a scale of 0 to 1.

Feature	Cramér's V	Association Strength	Significant?
Contract	0.410	**Strong**	Yes
InternetService	0.322	**Strong**	Yes
PaymentMethod	0.303	**Strong**	Yes
PaperlessBilling	~0.19	Moderate	Yes
OnlineSecurity	~0.18	Moderate	Yes
TechSupport	~0.18	Moderate	Yes
SeniorCitizen	~0.15	Moderate	Yes
Dependents	~0.16	Moderate	Yes
Partner	~0.15	Moderate	Yes
gender	~0.01	Weak	**No**
PhoneService	~0.01	Weak	**No**

Key conclusion: Contract type, internet service type, and payment method are the three strongest categorical predictors of churn. Gender and phone service are statistically insignificant.

Correlation Matrix (Numerical Features + Churn)

Pair	Correlation	Interpretation
tenure ↔ TotalCharges	+0.83	Long-tenure customers accumulate more total spend
tenure ↔ Churn	-0.35	Longer tenure = significantly lower churn
MonthlyCharges ↔ Churn	+0.19	Higher charges = slightly higher churn
TotalCharges ↔ Churn	-0.20	Higher total charges (long tenure) = slightly lower churn

Pair	Correlation	Interpretation
MonthlyCharges ↔ TotalCharges	+0.65	Higher monthly bills accumulate more total charges

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7. Multivariate Analysis

Contract × Internet Service Heatmap

Contract \ Internet	DSL	Fiber Optic	No Internet
Month-to-month	32.2%	**54.6%**	18.9%
One year	9.3%	19.3%	2.5%
Two year	1.9%	7.2%	0.8%

Insight: Month-to-month Fiber Optic customers have a 54.6% churn rate — the highest single segment. Two-year contracts are dramatically protective regardless of service type.

Contract × Payment Method Heatmap

Insight: Electronic check + month-to-month = 53.7% churn, the highest combination in this view. Auto-pay + two-year contracts consistently show the lowest churn (<4%).

Payment Method × Internet Service Heatmap

Insight: Fiber Optic + Electronic check = 53.2% churn. Automatic payment methods create "passive loyalty" — customers don't actively engage with billing, reducing impulse churn. No internet service customers are the most stable segment overall.

Churn Rate by Tenure Group & Contract Type

Tenure Group	Month-to-Month	One Year	Two Year
0–12 months	**~51%**	~12%	~Low
13–24 months	~38%	~10%	~Low
25–48 months	~32%	~9%	~Low
49–72 months	~26%	~8%	~Low

Insight: The first 12 months with a month-to-month contract is the highest-risk window. If a flexible-contract customer survives past 12 months, their churn probability drops substantially.

Number of Add-on Services vs Churn Rate

Services Count	Churn Rate
0	~Low
1	**~45%** (highest)

Services Count	Churn Rate
2	~Moderate
3+	Below average
6	~<10%

Insight: Customers with just 1 add-on service churn at the highest rate. As service count increases, switching costs rise and churn falls. Customers with 0 services (basic/core users) are actually more stable than 1-service customers.

High-Risk Segment Summary

Segment	Churn Rate
Month-to-Month + Fiber Optic	~54–55%
Month-to-Month + Electronic Check	~53–54%
Tenure 0–12 months + No Online Security	High
1 Add-on Service only	~45%
Senior Citizen + Month-to-Month	High

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8. Feature Engineering

Dropped Columns

Column	Reason
Churn (string)	Replaced by Churn_Binary (integer target)
tenure_group	EDA-only binned column, not needed for modelling

Engineered Features

Feature	Definition	Rationale
AvgMonthlySpend	TotalCharges / (tenure + 1)	Normalises spending intensity independent of tenure length; reduces multicollinearity
HighRisk_Flag	1 if Month-to-month AND Fiber Optic, else 0	Captures the highest-risk compound segment identified in multivariate analysis
HasSupportService	1 if OnlineSecurity OR TechSupport = Yes	Aggregates protective service subscriptions into a single flag
IsAutoPay	1 if payment is Bank transfer or Credit card (automatic)	Captures auto-pay loyalty signal

Feature	Definition	Rationale
num_services	Count of 6 add-on services subscribed	Quantifies ecosystem integration depth

Encoding

Column Type	Encoding Applied
Binary Yes/No columns (gender, Partner, Dependents, PhoneService, PaperlessBilling, SeniorCitizen)	Mapped to 0/1
Service columns (2-level: Yes/No)	Mapped to 0/1
Contract	Ordinal encoding: Month-to-month=0, One year=1, Two year=2
InternetService, PaymentMethod	One-hot encoding with drop_first=True

Scaling

Numerical features (tenure, MonthlyCharges, TotalCharges, AvgMonthlySpend, num_services) were standardised using StandardScaler — fit on training data only, applied to test data.

Final feature matrix shape: 7,043 rows × 27 features

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9. Modelling

Train-Test Split

- **Split ratio:** 80/20 (stratified by target to preserve class balance)
- **Train size:** ~5,634 rows
- **Test size:** ~1,409 rows
- **Churn rate in train/test:** ~26% (consistent due to stratification)

Models Trained

Logistic Regression

- Baseline linear model
- Hyperparameter: class_weight='balanced', max_iter=1000
- ROC-AUC: ~0.84

Random Forest

- Ensemble tree model
- Hyperparameters: n_estimators=200, max_depth=10, class_weight='balanced'
- ROC-AUC: ~0.83

XGBoost

- Gradient boosting model
- Hyperparameters: n_estimators=200, max_depth=5, learning_rate=0.05
- Class imbalance handled via scale_pos_weight = non-churners / churners
- ROC-AUC: 0.843

Model Comparison Summary

Model	Precision (Churn)	Recall (Churn)	F1 (Churn)	ROC-AUC
Logistic Regression	0.511	0.778	0.617	0.848
Random Forest	0.555	0.687	0.614	0.840
XGBoost	0.533	0.781	0.633	**0.843**

XGBoost was selected as the best model based on highest Precision, Recall, F1 and equivalent ROC-AUC.

ROC Curve

All three models were compared on the Precision, Recall, F1 and equivalent ROC-AUC.

XGBoost achieved highest amongst all.

Top 15 XGBoost Feature Importances

The model's built-in feature importances showed Contract, tenure, MonthlyCharges, TotalCharges, and the engineered AvgMonthlySpend and HighRisk_Flag as the top contributors.

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10. Threshold Optimisation

Why Threshold Matters

By default, classifiers predict "Churn" when probability ≥ 0.50. However, in a retention context, missing a churner (False Negative) is far more costly than a false alarm (False Positive). Lowering the decision threshold increases recall at the cost of some precision.

Precision-Recall-F1 Curve Analysis

The Precision-Recall-F1 curve was plotted across all thresholds. Two thresholds were identified:

Threshold	Objective	Result
Optimal F1 threshold	Maximise F1	Best balance of precision and recall
Business threshold (0.373)	Achieve ≥ 85% recall	**Selected for deployment**

Final Model Performance (Business Threshold = 0.373)

Metric	Value
ROC-AUC	0.843
Recall (Churn class)	**85.0%**
Precision (Churn class)	**47.8%**

Default vs Business Threshold Comparison

Metric	Default (0.50)	Business (0.373)
Recall (Churn)	~Lower	**85%**
Precision (Churn)	Higher	47.8%
Missed churners	More	**Fewer**
False alarms	Fewer	More

The business threshold catches significantly more churners, which is the primary goal for a retention campaign.

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11. Hyperparameter Tuning

Method

RandomizedSearchCV with 50 iterations and 5-fold Stratified CV was applied to the XGBoost classifier.

Search Space

Parameter	Values Searched
n_estimators	100, 200, 300, 500
max_depth	3, 4, 5, 6, 8
learning_rate	0.01, 0.05, 0.1, 0.2
subsample	0.6, 0.7, 0.8, 1.0
colsample_bytree	0.6, 0.7, 0.8, 1.0
min_child_weight	1, 3, 5
gamma	0, 0.1, 0.2, 0.5

Cross-Validation Results

Baseline XGBoost CV AUC was evaluated before tuning. After tuning, the improvement over the base model was marginal. The base XGBoost model with business threshold (0.373) was retained as the final model since tuning did not meaningfully increase recall or AUC.

12. SHAP Explainability

SHAP (SHapley Additive exPlanations) was used to explain model predictions at both global and local levels.

Global Explainability

Summary Plot (Dot)

Each point represents one customer. Position on the x-axis shows the direction and magnitude of impact on the churn prediction.

Key observations:

- **Contract:** Blue dots far left (two-year = strongly reduces churn); red dots near centre (month-to-month = moderate push toward churn)
- **Tenure:** Clear red-to-blue gradient — high tenure (red) is protective, low tenure (blue) increases risk
- **MonthlyCharges:** Red (high charges) pushes right (toward churn); wide spread indicates high variance in impact
- **HighRisk_Flag:** Binary split — clean, strong separation between flag=0 and flag=1

Bar Plot (Mean |SHAP|)

Ranks features by overall contribution:

1. Contract
2. tenure
3. MonthlyCharges
4. TotalCharges
5. HighRisk_Flag
6. AvgMonthlySpend
7. PaymentMethod_Electronic check
8. HasSupportService

Gender and num_services rank near the bottom, confirming they add minimal predictive value.

Dependence Plots (Top 4 Features)

Feature	Insight
Contract	Sharp 3-step function: month-to-month (+0.8 SHAP) → one-year (-0.8) → two-year (-2.5). Biggest single intervention available.
HighRisk_Flag	Binary split: 0 = ~-0.6 SHAP (protective), 1 = ~+0.3 (risky). Clean, consistent.

Feature	Insight
tenure	Smooth negative slope — protective effect is gradual and continuous, not threshold-based
MonthlyCharges	Low charges strongly protective (-2 SHAP); high charges moderately risky. Protective effect is larger than risk effect.

Local Explainability (Waterfall Plots)

Most Likely Churner

Profile of the customer with the highest predicted churn probability:

Feature	Value	SHAP Contribution
tenure	Very low	+0.93 (top churn driver)
Contract	Month-to-month	+0.86
TotalCharges	Low	+0.49
HighRisk_Flag	1	+0.34
MonthlyCharges	Above average	+0.26
PaymentMethod (Electronic check)	Yes	+0.23
HasSupportService	0	+0.16
SeniorCitizen	Yes	+0.15

Recommended intervention: Offer discounted annual contract with OnlineSecurity.

False Positive Analysis

A customer predicted to churn (93.1% probability) who actually stayed. Risk factors (contract, service type, tenure, charges) were all present, but two loyalty signals partially offset the prediction:

- Auto-pay method (not electronic check)
- Has support service subscription

Insight: This profile — high structural risk but with loyalty signals — is the ideal target for contract upgrade outreach. They are showing retention behaviour despite a high-risk profile.

Mean SHAP Difference: Churners vs Non-Churners

Feature	SHAP Difference
Contract	1.42 (largest differentiator)
tenure	0.67

Feature	SHAP Difference
HighRisk_Flag	0.30
MonthlyCharges	0.30
PaymentMethod_Electronic check	0.13
HasSupportService	0.06

All bars are positive — every top feature pushes more toward churn for churners than non-churners, confirming clean, consistent model learning.

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13. Business ROI Analysis

Assumptions

Input	Value
Average tenure of retained customers	Computed from data
Average monthly charges of churners	Computed from data
Estimated CLV per retained customer	Avg monthly charge × avg retained tenure
Cost per intervention	₹500
Assumed post-outreach retention rate	30%

Campaign Economics (30% Retention)

Metric	Result
Churners caught (85% recall)	~85% of actual churners in test set
False alarms	Determined by precision at business threshold
Total customers to contact	Caught + False alarms
Total campaign cost	Customers flagged × ₹500
Customers saved	Caught × 30%
Revenue protected	Customers saved × Estimated CLV
Net ROI	Revenue protected − Campaign cost
ROI multiple	Revenue protected ÷ Campaign cost

Sensitivity Analysis

Retention Rate	Customers Saved	Approx Net ROI
10%	Low	Likely negative
17%	Break-even	~₹0
20%	Moderate	Positive
30%	Good	Strongly positive
50%	High	Very strong

Key insight: The campaign breaks even at only ~17% retention. Even a modest intervention success rate produces meaningful positive ROI.

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14. Key Findings Summary

Highest-Risk Profile

Month-to-month contract + Fiber Optic internet + Electronic Check payment + tenure < 12 months

Three Strongest Churn Drivers

1. **Contract type** — single most powerful predictor (Cramér's V = 0.41, SHAP rank #1)
2. **Tenure** — first 12 months is the danger zone; churn drops sharply after this window
3. **Internet service type** — Fiber Optic customers consistently churn more than DSL or no-internet customers

Model Performance

- XGBoost achieved ROC-AUC of 0.843
- At business threshold of 0.373: 85% recall, 47.8% precision
- Campaign breaks even at 17% post-outreach retention rate

Feature Engineering Validation

- HighRisk_Flag (engineered) ranks in the top 5 SHAP features, confirming the Fiber Optic + month-to-month compound risk was a genuine signal worth capturing
- HasSupportService and IsAutoPay both appear in SHAP importance, validating those engineered flags

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15. Business Recommendations

1. Promote Long-Term Contract Adoption

Contract type is the single strongest churn lever. Month-to-month customers churn at 5–10× the rate of two-year customers.

Actions:

- Offer incentives for month-to-month customers to upgrade to 1-year or 2-year plans
- Example: "Switch to an annual plan and get 3 months at a discounted rate"
- Priority: New customers (0–12 month tenure) on flexible contracts

2. Strengthen Early-Customer Retention (First 12 Months)

The first year is the highest churn risk window, especially for month-to-month customers.

Actions:

- Proactive onboarding outreach in months 1–3
- Service adoption campaigns during this period
- Usage engagement programmes to build switching costs early

3. Encourage Auto-Pay Adoption

Electronic check users churn at significantly higher rates. Auto-pay creates passive loyalty.

Actions:

- Offer a small monthly discount for switching to automatic payment
- Simplify the payment migration process
- Target electronic check users specifically during retention campaigns

4. Expand Support-Service Adoption

Customers without OnlineSecurity or TechSupport churn more and have lower SHAP-measured loyalty scores.

Actions:

- Bundle OnlineSecurity or TechSupport with new Fiber Optic sign-ups
- Offer 3-month free trials during onboarding
- Target customers identified as HasSupportService = 0 in retention campaigns

5. Operationalise the Churn Model

Recommended workflow:

1. Score all active customers monthly using the XGBoost model at threshold 0.373
2. Prioritise the top-scored customers (highest churn probability) for retention outreach
3. Use SHAP waterfall plots to generate customer-specific intervention rationale
4. Retrain the model quarterly as customer behaviour patterns evolve

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16. Limitations

Limitation	Detail
Static snapshot	The dataset is a point-in-time extract; it does not capture customer behaviour changes over time
Missing external factors	Competitor pricing, regional factors, service outages, and marketing interactions were not available
Causality	The analysis identifies predictive associations, not causal relationships. Contract type correlates with churn, but the mechanism is not confirmed
Threshold assumptions	The business threshold (0.373) and ROI assumptions (₹500 cost, 30% retention rate) are illustrative and should be calibrated against actual campaign data
Model drift	Customer behaviour evolves; the model requires periodic retraining to remain effective

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Report prepared by: SRISHTI
Dataset: IBM Telco Customer Churn Dataset
Best-performing model: XGBoost (ROC-AUC: 0.843, Recall: 85% at threshold 0.373)